



DM270 ES1.1 Issue List

This document shows the issues found in the evaluation of DM270 ES1.1 (Rev 1.1), the workaround that should be applied in ES1.1, and correction that would be done in ES1.2. If a customer plans to use ES1.1 in his production system, then the customer should use the workarounds mentioned for ES1.1

1.

Issue	GIO0 function										
Description and workaround in ES1.1	In ES1.1 GIO0 is dedicated for WakeUp function from sleep or power-down mode. There is no change in functioning of GIO1-GIO33. Also the pin name is changed in ES1.1 as follows: <table border="1"> <tr> <td></td><td>ES1.0</td><td>ES1.1</td></tr> <tr> <td>Pin Name</td><td>GIO0</td><td>WAKEUP</td></tr> <tr> <td>Type</td><td>I/O</td><td>I (Input)</td></tr> </table> - Do not use GIO0 for general purpose I/O. Use only for wakeup of DM270 from power down or sleep mode - Setting of GIO0 to output direction (REG 0x30580, bit 0) should not be done - When GIO0 is not used for wakeup function, pull-up or tie the pin directly to GND			ES1.0	ES1.1	Pin Name	GIO0	WAKEUP	Type	I/O	I (Input)
	ES1.0	ES1.1									
Pin Name	GIO0	WAKEUP									
Type	I/O	I (Input)									
Correction in ES1.2	The same workaround as ES1.1, should also be done in ES1.2										

2.

Issue	SDRAM Clock Enable Specification	
Description and workaround in ES1.1	In ES1.1, default (power-on reset) state of SDRAM clock depends on FLSHSEL signal - when FLSHSEL=0, SDRAM clock is disabled - when FLSHSEL=1, SDRAM clock is enabled Also automatic setting of SDRAM clock at the time of wakeup is performed only when FLSHSEL=1 Thus, in ES1.1, when DM270 is activated from sleep or power down mode using WakeUp function and FLSHSEL=Low, SDRAM clock will be always be disabled.	
Correction in ES1.2	SDRAM clock state after wakeup from sleep or power down would operate same as other peripherals.	

3.

Issue	PLL Output Clock Frequency	
Description and workaround in ES1.1	In ES1.0, PLL output clock frequency restriction is: $200 \text{ MHz} < F_{in} * (M/N) < 351 \text{ MHz}$ In ES1.1 and higher, PLL output clock frequency restriction is $110 \text{ MHz} \leq F_{in} * (M/N) \leq 351 \text{ MHz}$ See also “ARM clock divider input frequency restriction”	
Correction in ES1.2	The same restriction as ES1.1, is applicable for ES1.2	

4.

Issue	ARM clock divider input frequency restriction
Description and workaround in ES1.1	In ES1.0 and ES1.1 input clock to ARM clock divider should be less than 203 Mhz.
Correction in ES1.2	In ES1.2, input clock to ARM clock divider should be less than 351 Mhz.

5.

Issue	Default ARM clock divider input frequency
Description and workaround in ES1.1	In ES1.1, default PLL clock frequency input to ARM clock divider is 243Mhz. But due to the restriction in ES1.1 this input clock frequency must be less than 203Mhz When ARM is not in external host boot mode (EXTHOST=Low), PLL is bypassed, so user should set appropriate value in PLL before switching ARM clock input to PLL clock But, when ARM is in external host boot mode (EXTHOST=High), PLL is not bypassed, hence input to ARM clock divider becomes 243Mhz. This could cause damage to the DM270. Hence DM270 ES1.1 should not be used in external host mode NOTE: This assumes that the input clock frequency to PLL is 27Mhz
Correction in ES1.2	Default PLL output clock frequency would be changed from 243Mhz to 121.50Mhz, in ES1.2. So DM270 ES1.2 can be used in external host mode

6.

Issue	Default PLL Output Clock Frequency
Description and workaround in ES1.1	In ES1.1, default PLL output clock frequency is 243Mhz. Due to this, default clock frequency of DSP is also 243Mhz.. The user should make sure the proper DSP clock frequency is set before enabling clock to DSP
Correction in ES1.2	Default PLL output clock frequency would be changed from 243Mhz to 121.50Mhz However, the user should still make sure the proper DSP clock frequency is set before enabling clock to DSP

7.

Issue	External Interruption to DSP
Description and workaround in ES1.1	In ES1.1, DSP can be interrupted externally via GIO6. But this interrupt detection is level signal detection The minimum pulse width detected as an interrupt is "2 DSP clocks with low level".
Correction in ES1.2	In ES1.2, this interrupt detection would be edge signal detection

8.

Issue	DSP ROM Code of MemoryRdFast_NonWait(), MemoryRdFast_Wait(), MemoryWrFast_NonWait(), MemoryWrFast_Wait(), MemoryAccessWait(), SDRAMRdFast(), SDRAMWrFast(), SDRAMDualRdFast()
Description and workaround in ES1.1	This DSP ROM code cannot be used due to a bug in the ROM code. The routines in DSP ROM are used for image buffer DMA. When image buffer DMA completes, image buffer DMA interrupt is cleared by the ROM code. But due to the bug in the ROM code, all pending interrupts will also be cleared. This can cause problems in a system where multiple interrupts are being used. The user has two workaround options: 1. In case he is not using interrupts in his S/W on the DSP side then he can use the APIs which are present in the ROM. 2. In case he is using interrupt based S/W on the DSP side, then the user must use the program RAM to store these routines. The source code of these routines is available from TI
Correction in ES1.2	The same workaround as ES1.1, should also be done in ES1.2

9.

Issue	Interrupt to SDIO-IF without clock
Description and workaround in ES1.1	When the SDIO-IF is used in "SD 1-bit mode" and the clock "SD_CLK" is stopped, DM270 ES1.1 can not recognize any interrupt through DAT[1]. The user has two workaround options: 1. Do not turn off SD_CLK in 1-bit mode. The clock is enabled by CLKEN bit in MMCCLK register(0x003:0482). 2. If SD_CLK need to be turned off, connect a GIO(1 to 15) and SD_DATA[1] physically and use the GIO as an external interrupt input. When SD_CLK is enabled, disable the GIO interrupt and enable the SDIO interrupt. When SD_CLK is disabled, enable the GIO interrupt and disable the SDIO interrupt by software.
Correction in ES1.2	The same workaround as ES1.1, should also be done in ES1.2

10.

Issue	Unexpected USB DMA Interrupt Assertion
Description and workaround in ES1.1	In ES1.1 or ES1.2, user will unexpectedly get a USB DMA interrupt, MIDMA (REG 0x30B0A, bit15), when a short (or null) packet is received on an OUT endpoint. This occurs only when DMA target endpoint register, DEP (REG 0x30B18, bit14-12), is set to "000" or "010". In case of "000", a short packet reception onto EP0 OUT causes a DMA interrupt. In case of "010", a short packet reception onto EP1 OUT causes a DMA interrupt.. It occurs regardless of whether DMA transmission is started or not. Workaround: Ignore USB DMA interrupt under the following conditions; 1. EP0 OUT short packet reception with DEP="000" 2. EP1 OUT short packet reception with DEP="010"
Correction in ES1.2	The same workaround as ES1.1, should also be done in ES1.2

11.

Issue	HPIB DMA Interrupt Lost
Description and workaround in ES1.1	HPIB DMA asserts an interrupt to DSP at DMA transfer completion. But there might be a rare occasion where the interrupt would be lost. It is uncertain when this interrupt is lost. Workaround: Use polling of DMA status bit (REG DMATRG 0x0086, bit0) to detect HPIB DMA completion. Do not use HPIB DMA interrupt.
Correction in ES1.2	The same workaround as ES1.1, should also be done in ES1.2

12.

Issue	HPIB DMA Abort Malfunction
Description and workaround in ES1.1	HPIB DMA provides abort function by writing 0 to bit0 of DMATRG (0x86) register. However, if another DMA transfer is started after abort, a deadlock might happen. Applying system reset is the only way to recover from this deadlock Workaround: Do not use HPIB DMA abort function. Please make sure to wait for the DMA completion once DMA is started.
Correction in ES1.2	The same workaround as ES1.1, should also be done in ES1.2

NOTE:

All DM270 ES1.0 issues listed in the DM270 ES1.0 Issue list rev 1.2 document (24 Mar 2003) and DM270 ES1.1 Enhanced Function rev1.0 document (14 Mar 2003), have been fixed in DM270 ES1.1, **EXCEPT** for the issues listed below:

1. #31 "OSD : Band of irregular filter" : This issue remains in both ES1.1 and ES1.2
2. #20 "USB : UDC bug " : This issue has been fixed in ES1.1, but evaluation with EVM is not yet completed